

Expectations

- **Definition:** If $X: \Omega \rightarrow D$ is a **discrete r.v. (DRV)** and $f: D \rightarrow R$ is a value function, then the **expectation of $f(X)$** is defined as

$$E[f(X)] = \sum_{i \in D} f(i) P\{X = i\}$$

- **Remark:** if $D \subset R$ then we can choose $f(i) = i$ and the expectation of X becomes:

$$E[f(X)] = \sum_{i \in D} i P\{X = i\}$$

- **Note:** this definition makes sense **only for DRV**, because for **continuous r.v. (CRV)** we have $P\{X = x\} = 0$, for all $x \in D$.

Expectations

- **Example 1:**

- Consider the DRV: $X = \text{"suit of a card in a deck"}$
- **Modeling assumptions:** Each suit is equally likely $\rightarrow P\{X = \text{"any suit"}\} = 1/4$
- The **value function** $f: D \rightarrow \{1, 2, 3, 4\}$, where $D = \{\spadesuit, \diamondsuit, \clubsuit, \heartsuit\}$
 - Think of the values 1, 2, 3, 4 as **the number of \$\$\$ you get if you get that suit**

- Then: $E[f(X)] = \sum_{x \in D} f(x) P\{X = x\} =$

$$f(\spadesuit)P\{X = \spadesuit\} + f(\diamondsuit)P\{X = \diamondsuit\} + f(\clubsuit)P\{X = \clubsuit\} + f(\heartsuit)P\{X = \heartsuit\} =$$

$$= 1\frac{1}{4} + 2\frac{1}{4} + 3\frac{1}{4} + 4\frac{1}{4} = \frac{10}{4} = \frac{5}{2}.$$

Expectations – *Indicator Function*

- **Example 2:** (very IMPORTANT for the rest of course!)

- Assume A is an event. Then we can define the DRV $\mathbf{1}_A$ with values:

$$1_A(\omega) = \begin{cases} 1, & \text{if } \omega \in A \\ 0, & \text{if } \omega \notin A \end{cases}$$

- The DRV is called the indicator RV.
- Let's compute the **expectation** of 1_A

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- Let's compute the **expectation** of 1_A

$$E[1_A] = 1P\{1_A = 1\} + 0P\{1_A = 0\} = P\{1_A = 1\} = P(A)$$

- So, the expectation of $\mathbf{1}_A$ is the probability the event A

Indicator Function – Usefulness!

The indicator function allows us to represent the **number of successes in a series of trials** as a simple sum.

Example:

- Consider the DRV X representing the **number of heads** in n coin flips (**Successes**).
- Define the **events**: $X_i = \text{"the } i\text{-th flip is head"}$ for $i = 1, 2, \dots, n$. (**i -th success**)
- For each event X_i we can define the corresponding **indicator function** 1_{X_i} (**i -th success or failure**)
- Then we can express X as the sum of the indicator function:

$$X = 1_{X_1} + 1_{X_2} + \dots + 1_{X_n}$$

Recap from previous Lecture

- **Definition:** If $X: \Omega \rightarrow D$ is a **DRV** and $f: D \rightarrow R$ is a function, then the **expectation of X** is

$$E[f(X)] = \sum_{i \in D} f(i)P\{X = i\}$$

- **Definition:** If $X: \Omega \rightarrow D$ is a DRV, then its **distribution** is given by specifying the probabilities of all its outcomes $(P\{X = i\})_{i \in D}$
- **Definition:** the **Joint Distribution** of two DRVs $X: \Omega \rightarrow D_X$ and $Y: \Omega \rightarrow D_Y$ is specified by all probabilities $(P\{X = i, Y = j\})_{i \in D_X, j \in D_Y}$
- **Definition:** **Marginal Distribution** (when we know the Joint): $P\{X = i\} = \sum_{j \in D_Y} P\{X = i, Y = j\}$

Expectation – Joint and Marginal Distributions

- Example:** You know the distributions of 2 DRV's X, Y and want to compute $E[X + Y] = ??$

$$E[X + Y] = \sum_{x \in D_X} \sum_{y \in D_Y} (x + y) P\{X = x, Y = y\} =$$

$$= \sum_{x \in D_X} \sum_{y \in D_Y} x P\{X = x, Y = y\} + \sum_{x \in D_X} \sum_{y \in D_Y} y P\{X = x, Y = y\} =$$


$$\sum_{x \in D_X} x \underbrace{\sum_{y \in D_Y} P\{X = x, Y = y\}}_{\text{Marginal of } X \text{ (sum over } y\text{'s)}} + \sum_{y \in D_Y} y \underbrace{\sum_{x \in D_X} P\{X = x, Y = y\}}_{\text{Marginal of } Y \text{ (sum over } x\text{'s)}} = \sum_{x \in D_X} x P\{X = x\} + \sum_{y \in D_Y} y P\{Y = y\} =$$

$$= E[X] + E[Y]$$

Indicator function & Expectation

Indicator function and Expectations

- **Example:**

- Random Experiment: Flip n **identical** coins. You are interested in the **total number of Heads** that may come up, **on average**, if you repeat the experiment many times.
- Define the DRV: $X = \text{"the total \# of H"}$
- **Question**: What is $E[X] = ??$
 - Define the **events**: $A_i = \text{"the } i\text{-th coin shows H"}$, $i = 1, 2, \dots, n$
 - $P\{A_i\} = p$ and $P\{A_i^c\} = 1 - p$

Indicator function and Expectations

- **Example (cont.):**

- Define the indicator function: $1_{A_i} = \begin{cases} 1, & \text{if } A_i = \text{True} \\ 0, & \text{otherwise} \end{cases}$
- Then we can write $X = 1_{A_1} + 1_{A_2} + \cdots + 1_{A_n}$
- We have: $E[X] = E[1_{A_1} + 1_{A_2} + \cdots + 1_{A_n}] = E[1_{A_1}] + E[1_{A_2}] + \cdots + E[1_{A_n}] =$
 $= P\{A_1\} + P\{A_2\} + \cdots + P\{A_n\} = np$

Indicator function and Expectations

- **Example (cont.):**

- **Note:** we did **NOT** have to assume that $A_i \perp\!\!\!\perp A_j, \forall i \neq j$. This means that even if we were **not** flipping the coins independently (but through a **collusion or conspiracy**) we would have gotten the same result: $E[X] = np$
 - A possible **collusion** or **conspiracy**: "I flip my coin and then I force everyone to turn their coins to my outcome".
- **Note:** We will often compute expectations using **tricks** (i.e., using indicator functions) similar with those in this example.